Apache Kafka Internals: Exactly-Once Processing

Idempotent Producers, Transactional Coordinators & Consumer Isolation

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1. Idempotent Producer Mechanics

PID [Producer ID] assignment via InitProducerIdRequest

Monotonically increasing sequence numbers per TopicPartition

Broker deduplication within current producer session window

2. Transactional Coordinator Protocol

Two-phase commit over internal __transaction_state topic

Control markers [COMMIT / ABORT] appended directly to partition log

Consumer read_committed isolation filters non-committed batches